BANK LOAN ANALYSIS

Professional Data Analyst Project Report

End-to-End Finance Domain Analytics Project

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Purpose of this report

This report documents the business problem, data, analytical workflow, KPI definitions, SQL analysis, Python analysis, Power BI reporting, business interpretation, recommendations, limitations, and interview-ready explanation of the Bank Loan Analysis project.

1. Executive Summary

The Bank Loan Analysis project is an end-to-end data analytics solution for monitoring lending activity and loan portfolio performance. The project uses Python for initial analysis, MS SQL Server for structured business analysis and validation, and Power BI for interactive reporting.

The source analysis contains 38,576 records and 24 columns. The dataset includes loan, borrower, risk, payment and date attributes such as loan status, grade, sub-grade, annual income, DTI, interest rate, loan amount and total payment.

The reporting layer is organized into three dashboards: Summary, Overview and Details. The Summary dashboard focuses on executive KPIs and Good Loan versus Bad Loan performance. The Overview dashboard provides trend and segmentation analysis. The Details dashboard provides record-level information for investigation.

The documented business requirements emphasize total applications, funded amount, amount received, average interest rate, average DTI, MTD and MoM analysis, loan status, monthly trends, state analysis, loan term, employee length, purpose and home ownership.

A key portfolio result from the Python analysis is 33,243 Good Loan applications, representing 86.18%, versus 5,333 Bad Loan applications, representing 13.82%.

2. Business Context and Problem Statement

The business objective is to monitor and assess bank lending activities and performance, understand changes over time, track portfolio health, and identify trends that can inform lending strategies.

The business requirement specifically asks for a comprehensive Bank Loan Report rather than a simple data table. The report therefore combines high-level management KPIs with analytical segmentation and detailed loan-level information.

The analysis supports decisions around portfolio monitoring, lending performance, risk assessment, borrower segmentation and cash-flow visibility. Domain documentation also describes DTI, income verification, employment verification, risk assessment, loan approval, disbursement, repayment and ongoing monitoring as relevant lending concepts.

The key business questions are: How many applications are being received? How much is being funded? How much is being collected? What is the average interest rate? What is borrower DTI? What proportion is Good versus Bad? Which states, purposes, terms, employment groups and home-ownership groups drive activity?

3. Business Requirements Document Translation

Requirement	Analytical interpretation
Total Loan Applications	Count of loan/application records received during the selected period.
MTD Applications	Applications received from the start of the current month through the selected date
MoM Applications	Percentage change in applications compared with the prior month.
Total Funded Amount	Sum of loan amounts disbursed to borrowers.
Total Amount Received	Sum of borrower payments received.
Average Interest Rate	Average interest rate across loans in the selected context.
Average DTI	Average debt-to-income ratio of borrowers in the selected context.
Good Loan KPIs	Measures based on Fully Paid and Current statuses.
Bad Loan KPIs	Measures based on Charged Off status.
Loan Status Grid	Status-level comparison of applications, funding, collections, rate and DTI.
Overview Analysis	Monthly, state, term, employee length, purpose and home ownership analysis.
Details Dashboard	Detailed record-level view of loan attributes.

4. Data Source and Dataset Profile

The Python analysis documents a dataset of 38,576 rows and 24 columns. The fields cover borrower identifiers, geographic attributes, employment, loan characteristics, risk classification, dates, income, DTI, installments and payment information.

Area	Examples from dataset
Identifiers	id, member_id
Borrower	annual_income, emp_length, emp_title, home_ownership
Loan	loan_amount, term, purpose, installment
Risk	grade, sub_grade, dti, verification_status
Status	loan_status
Dates	issue_date, last_payment_date, next_payment_date, last_credit_pull_date
Payment	total_payment
Geography	address_state
Account history	total_acc

The documented data types include integer, string/object, datetime and float fields. Issue dates and payment dates are represented as datetime fields in the Python analysis.

5. Data Dictionary and Metric-Relevant Fields

Field	Meaning in project
id	Unique loan/application identifier used for counting records.
loan_status	Current state of the loan, used for Good/Bad classification.
loan_amount	Principal amount funded/disbursed.
total_payment	Total amount received from borrowers.
int_rate	Interest rate associated with the loan.
dti	Debt-to-income ratio used as a borrower financial-health indicator.
issue_date	Loan issue date used for time analysis.
address_state	Borrower state used for regional analysis.
term	Loan repayment term such as 36 or 60 months.
emp_length	Borrower employment-length category.
purpose	Stated purpose of the loan.
home_ownership	Borrower housing status such as RENT, MORTGAGE or OWN.
grade	Loan risk grade.
sub_grade	More granular loan risk classification.
installment	Scheduled periodic loan installment.
annual_income	Borrower annual income.

6. Python Analysis Layer

Python is used as the initial analytical layer. The documented workflow includes importing the dataset, inspecting structure, reviewing data types and descriptive statistics, analyzing loan categories and preparing the data for downstream SQL and BI reporting.

Typical analytical actions in the project include loading the data into a pandas DataFrame, converting issue_date to datetime, filtering Good and Bad loan records, calculating aggregate metrics and preparing time-based calculations.

The project documentation identifies pandas and NumPy as data-analysis libraries and includes visualization-oriented libraries such as Matplotlib, Seaborn and Plotly in the Python environment.

Example analytical logic for Good Loans is: filter loan_status to Fully Paid or Current, count the resulting records, sum loan_amount, sum total_payment and divide the Good Loan count by total application count to obtain the Good Loan percentage.

This Python layer provides a transparent analytical checkpoint before the final Power BI dashboard.

7. Data Preparation, Validation and Quality Checks

Data preparation is an important control step because dashboard metrics are only useful when the underlying fields are correctly typed, consistently aggregated and aligned with the business definition.

• Validate row count and column count before analysis.

• Inspect data types, especially date, numeric and categorical fields.

• Convert issue_date to a proper datetime field for month and MTD analysis.

• Check loan_status values before defining Good and Bad Loan groups.

• Confirm loan_amount and total_payment are numeric before aggregation.

• Validate DTI and interest-rate values before calculating averages.

• Compare Power BI totals with SQL results to identify calculation or filter discrepancies.

• Test dashboard results under slicer selections such as Grade A.

The project SQL documentation explicitly recommends modifying queries to validate dashboard results under filters, for example by applying WHERE grade = 'A'.

8. SQL Server Analysis Layer

MS SQL Server is used as the structured database and business-analysis layer. The project documentation covers database creation, table creation, SELECT statements, date functions, grouping, ordering, COUNT, DISTINCT, CTEs, partitions and other SQL concepts.

The SQL layer answers business questions using COUNT for application volume and SUM for funded and received amounts. GROUP BY is used to segment results by term, employee length, purpose and home ownership.

Example: Total applications can be calculated using COUNT(id). Total funded amount can be calculated using SUM(loan_amount). Total amount received can be calculated using SUM(total_payment).

For Good Loans, the business definition is applied through a status filter: Fully Paid and Current. For Bad Loans, the status is Charged Off.

SQL also provides an independent validation layer for Power BI. This is particularly useful when checking totals after slicer selections.

9. KPI Framework — Executive Metrics

Metric	Definition	Why it matters
Total Loan Applications	COUNT of loan/application records.	Measures demand and lending activity vo
MTD Applications	Applications from month start to selected date.	Shows current-month pace.
MoM Applications	Current month compared with previous month.	Shows growth or decline.
Total Funded Amount	SUM of loan_amount.	Shows capital deployed.
MTD Funded Amount	Current-month funded amount.	Shows current funding pace.
Total Amount Received	SUM of total_payment.	Shows borrower collections/cash inflow.
MTD Amount Received	Current-month collections.	Shows current collection pace.
Average Interest Rate	Average of int_rate.	Shows portfolio pricing level.
Average DTI	Average of dti.	Provides borrower debt-burden context.

10. KPI Explanation with Numerical Examples

Example 1 — Total Loan Applications: If the dataset contains 1,000 loan records after applying the selected filters, Total Loan Applications = 1,000.

Example 2 — MTD Applications: If 85 applications were issued from the first day of the month through the reporting date, MTD Applications = 85.

Example 3 — MoM: If the current month has 1,100 applications and the previous month had 1,000, MoM growth = (1,100 – 1,000) / 1,000 × 100 = 10%.

Example 4 — Funded Amount: If three loans have funded amounts of $10,000, $15,000 and $25,000, Total Funded Amount = $50,000.

Example 5 — Amount Received: If borrower payments received are $11,000, $16,500 and $27,000, Total Amount Received = $54,500.

Example 6 — Average Interest Rate: For rates of 10%, 12% and 14%, the simple average is (10 + 12 + 14) / 3 = 12%.

Example 7 — Average DTI: For DTI values of 10%, 15% and 20%, the simple average is 15%. A lower DTI generally indicates lower debt burden relative to income.

11. Good Loan vs Bad Loan Metrics

The project defines Good Loans as loans with status Fully Paid or Current. Bad Loans are loans with status Charged Off.

Metric	Formula / logic	Project result
Good Loan Applications	COUNT where status = Fully Paid or Current	33,243
Good Loan %	Good applications / total applications × 100	86.18%
Good Loan Funded Amount	SUM loan_amount for Good Loans	$370.22M
Good Loan Received	SUM total_payment for Good Loans	$435.79M
Bad Loan Applications	COUNT where status = Charged Off	5,333
Bad Loan %	Bad applications / total applications × 100	13.82%
Bad Loan Funded Amount	SUM loan_amount for Bad Loans	$65.53M
Bad Loan Received	SUM total_payment for Bad Loans	$37.28M

Example: If 1,000 applications exist and 850 are Fully Paid/Current, Good Loan % = 850/1,000 × 100 = 85%.

12. Loan Status Analysis

The Summary dashboard includes a Loan Status grid so that management can compare portfolio performance across statuses.

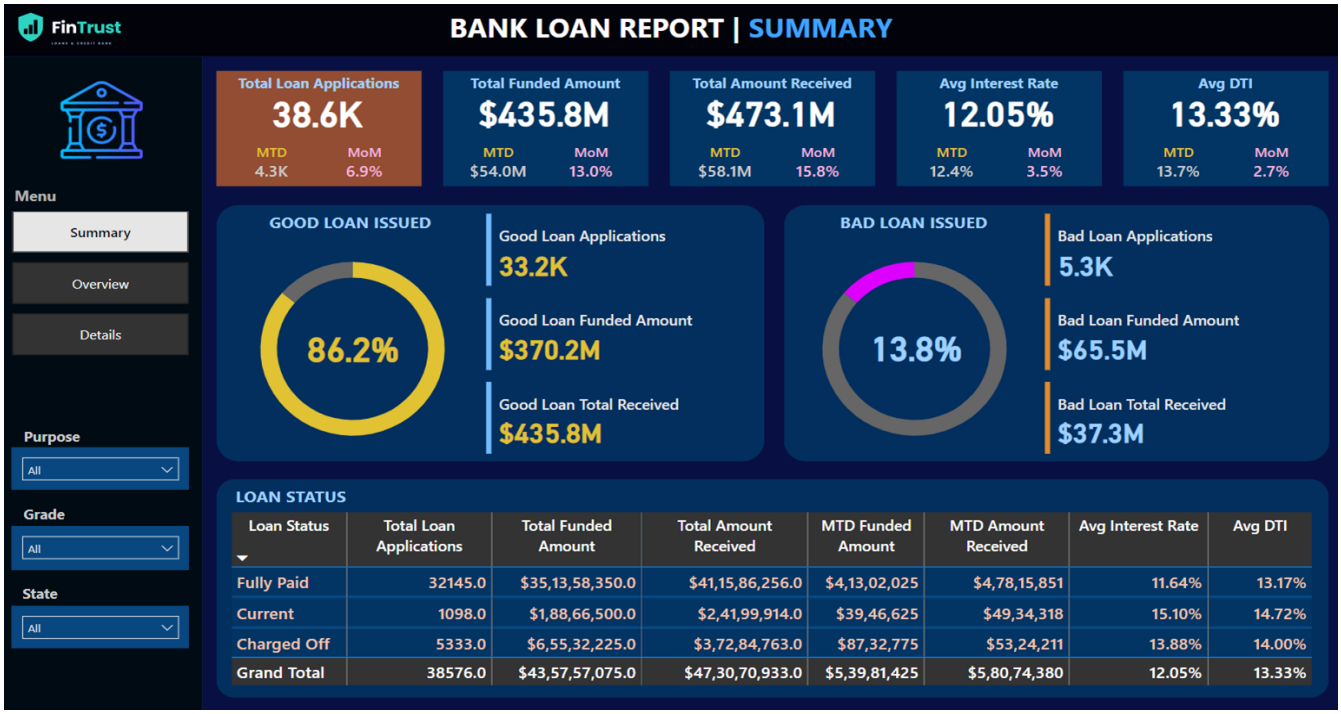
Status	What to examine
Fully Paid	Completed repayment performance and received amount.
Current	Loans currently active and continuing through repayment.
Charged Off	Loans classified as bad under the project definition.
Grand Total	Overall portfolio benchmark.

The status grid contains Total Loan Applications, Total Funded Amount, Total Amount Received, MTD Funded Amount, MTD Amount Received, Average Interest Rate and Average DTI.

Interview interpretation: a status-level table is useful because portfolio totals can hide differences between healthy and stressed segments. Status analysis helps the analyst move from 'how much business do we have?' to 'what is the quality and performance of that business?'.

13. Summary Dashboard — Executive View

The Summary dashboard is the executive reporting layer. It combines five top-level cards, Good Loan versus Bad Loan visuals and the Loan Status grid.



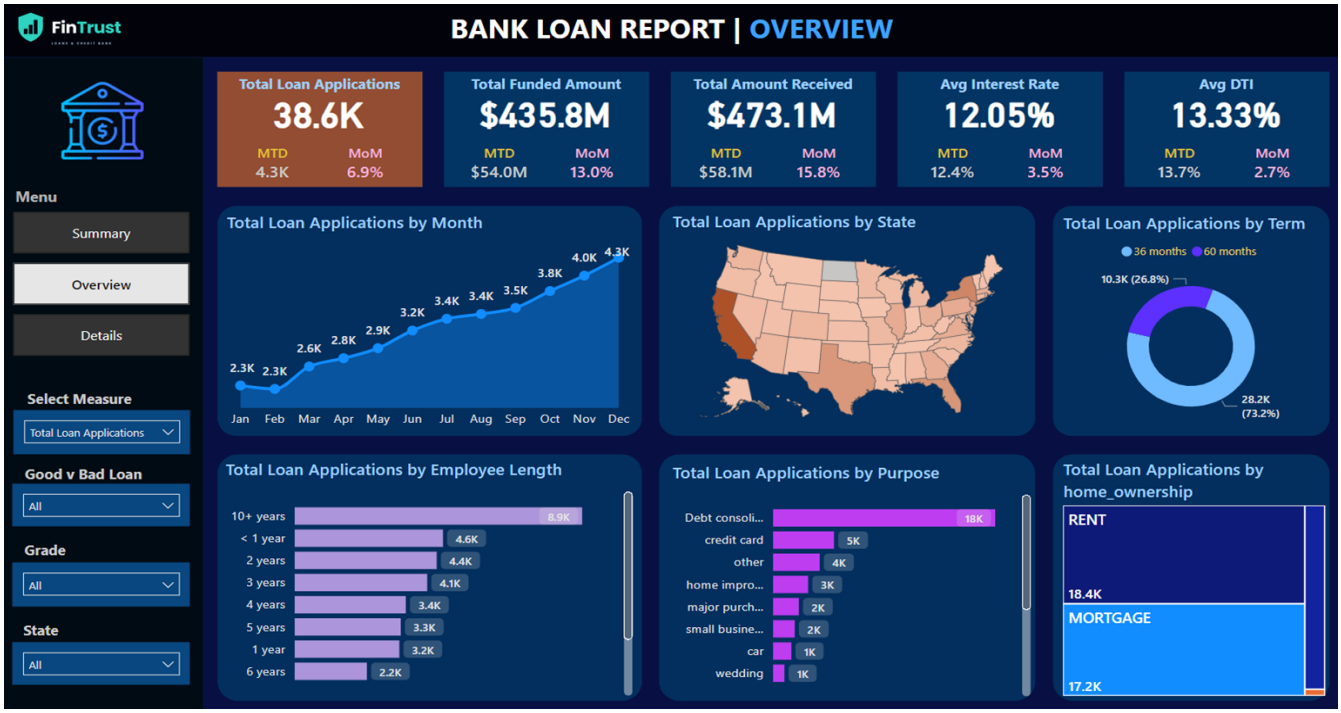
Loan Status	Total Loan Applications	Total Funded Amount	Total Amount Received	MTD Funded Amount	MTD Amount Received	Avg Interest Rate	Avg DTI
Fully Paid	32145.0	$35,13,58,350.0	$41,15,86,256.0	$4,13,02,025	$4,78,15,851	11.64%	13.17%
Current	1098.0	$1,88,66,500.0	$2,41,99,914.0	$39,46,625	$49,34,318	15.10%	14.72%
Charged Off	5333.0	$6,55,32,225.0	$3,72,84,763.0	$87,32,775	$53,24,211	13.88%	14.00%
Grand Total	38576.0	$43,57,57,075.0	$47,30,70,933.0	$5,39,81,425	$5,80,74,380	12.05%	13.33%

The dashboard shown above presents the portfolio-level KPIs and the Good/Bad Loan split. It also exposes MTD and MoM values to provide a current-period comparison.

Professional explanation: 'I designed the Summary page for management consumption. The first row answers the five most important portfolio questions, while the middle section evaluates loan quality and the bottom grid provides status-level detail.'

14. Overview Dashboard — Trend and Segmentation View

The Overview dashboard expands the analysis beyond headline KPIs. It contains monthly application trends, state-level analysis, term distribution, employee-length analysis, purpose analysis and home-ownership analysis.



This page helps answer 'where, when and for whom is lending activity occurring?' rather than only reporting the overall portfolio size.

15. Details Dashboard — Record-Level Analysis

The Details dashboard provides a detailed grid for users who need to move from aggregated KPIs to individual loan records.



id	purpose	home_ownership	grade	sub_grade	issue_date	Funded Amount	Int Rate	Sum of installment	Amount Collection
54734	Debt consolidation	RENT	B	B4	09 August 2021	$25,000.0	11.89%	$829.1	$29,330.0
55742	credit card	RENT	B	B5	08 May 2021	$7,000.0	10.71%	$228.2	$8,216.0
57245	Debt consolidation	OWN	C	C2	10 March 2021	$1,200.0	13.11%	$40.5	$1,458.0
57416	Debt consolidation	RENT	C	C3	09 November 2021	$10,800.0	13.57%	$366.9	$13,208.0
58915	Debt consolidation	RENT	B	B3	08 April 2021	$7,500.0	10.08%	$162.3	$5,844.0
59006	credit card	MORTGAGE	C	C5	09 September 2021	$3,000.0	14.26%	$102.9	$3,705.0
61390	credit card	MORTGAGE	A	A5	10 February 2021	$4,000.0	7.88%	$125.1	$4,452.0
61419	Debt consolidation	RENT	D	D2	10 February 2021	$5,600.0	14.96%	$194.0	$6,475.0
62102	Debt consolidation	RENT	B	B1	10 April 2021	$3,200.0	9.88%	$103.1	$3,414.0
65426	car	MORTGAGE	B	B1	09 August 2021	$4,000.0	11.14%	$131.2	$2,755.0
65640	home improvement	MORTGAGE	C	C2	08 May 2021	$5,000.0	11.34%	$87.2	$3,154.0
66431	Debt consolidation	RENT	B	B5	09 February 2021	$2,525.0	12.21%	$84.1	$3,028.0
66749	Debt consolidation	MORTGAGE	C	C4	08 December 2021	$10,625.0	13.47%	$360.4	$12,975.0
66943	Debt consolidation	RENT	B	B4	10 August 2021	$2,800.0	11.49%	$61.6	$3,144.0
66964	Debt consolidation	MORTGAGE	D	D3	08 June 2021	$7,500.0	13.24%	$253.6	$9,129.0
67503	Debt consolidation	MORTGAGE	A	A4	09 October 2021	$10,000.0	8.59%	$316.1	$11,280.0
68163	small business	MORTGAGE	A	A3	10 February 2021	$3,000.0	7.14%	$92.8	$3,342.0
68381	Debt consolidation	RENT	A	A5	08 March 2021	$6,625.0	8.63%	$209.5	$7,542.0

Displayed attributes include loan ID, purpose, home ownership, grade, sub-grade, issue date, funded amount, interest rate, installment and amount collection.

This page is particularly useful for validation, investigation and drill-down. It allows a user to inspect the underlying records that contribute to aggregated results.

16. Trend Analysis and Time Intelligence

Monthly trend analysis uses Issue Date to show changes in application volume over time. The business requirement specifically calls for a line chart so that seasonality and long-term trends can be identified.

Example interpretation: if applications rise from 2,300 in one month to 4,300 in another, the analyst should not stop at reporting the increase. The next question is why the increase occurred and whether funded amount and received amount increased proportionately.

MTD is a partial-month measure. It must be compared with the same type of time context and should not be confused with a full-month total.

MoM is a relative comparison. Formula: (Current Month − Previous Month) / Previous Month × 100. For example, 4,300 versus 4,000 gives 7.5% growth.

The Power BI documentation includes date tables and time-intelligence functions as part of the project learning scope.

17. Segmentation Analysis

State Analysis: identifies geographic concentration and regional differences in lending activity.

Loan Term Analysis: compares 36-month and 60-month terms and helps understand the structure of the loan book.

Employee Length Analysis: compares lending activity across borrower employment-length groups and can be used as a segmentation lens.

Purpose Analysis: identifies the reasons borrowers seek financing. Debt consolidation, credit card, home improvement, major purchase, small business, car and wedding are among the purposes represented in the dashboard.

Home Ownership Analysis: compares groups such as RENT, MORTGAGE and OWN. The SQL documentation specifically groups applications, funded amount and received amount by home ownership.

A professional analyst should avoid claiming causation from these segments. A difference between two groups indicates an association in the observed data, not necessarily that one variable caused the outcome.

18. SQL-to-Power BI Validation Framework

A strong Data Analyst workflow does not assume that a Power BI visual is correct simply because it displays a number. The project includes SQL queries that can independently reproduce core metrics.

Validation process:

• Start with a known business definition.

• Write the equivalent SQL aggregation.

• Check the Power BI measure under the same filter context.

• Compare counts and monetary totals.

• Apply a slicer such as Grade A and rerun the SQL with WHERE grade = 'A'.

• Investigate any mismatch before publishing the dashboard.

For example, the SQL documentation groups purpose by total applications, funded amount and received amount and demonstrates how a Grade A filter can be added to reproduce a filtered dashboard view.

19. Business Insights from the Project

1. Good Loan share is high: 33,243 applications are classified as Good Loans, representing 86.18% of applications.

2. Bad Loans remain material: 5,333 applications, or 13.82%, are classified as Charged Off. This makes Bad Loan monitoring important even though it is the smaller share.

3. Good Loans generated substantially more received amount than Bad Loans in the documented analysis: approximately $435.79M versus $37.28M.

4. Monthly application activity is important for understanding lending momentum. The Overview dashboard is designed to reveal changes over time.

5. Purpose, geography, term, employment length and home ownership provide useful segmentation dimensions for portfolio analysis.

6. Interest rate and DTI provide risk/pricing context alongside volume and financial measures.

These are descriptive insights from the project data and should not be interpreted as causal findings without additional statistical analysis.

20. Business Recommendations

Recommendation 1 — Monitor Bad Loan concentration: Track Charged Off applications and funded amounts by month, state, purpose, grade and other available dimensions.

Recommendation 2 — Build an early-warning view: Combine DTI, grade/sub-grade, employment length and repayment status for more focused portfolio monitoring.

Recommendation 3 — Monitor collection performance: Compare funded amount with amount received and observe how this relationship changes by status and segment.

Recommendation 4 — Investigate regional concentration: Large application or funding volumes in specific states can justify deeper risk and performance analysis.

Recommendation 5 — Use monthly trend monitoring: MTD and MoM measures should be reviewed regularly to identify sudden changes in demand, funding or collections.

Recommendation 6 — Validate before publishing: Maintain SQL-to-Power BI reconciliation as a standard quality-control step.

These recommendations are analytical actions based on the available dashboard dimensions; they are not underwriting policy decisions.

21. Limitations and Responsible Interpretation

The project is primarily descriptive and diagnostic. It summarizes historical loan data and does not, by itself, provide a predictive default model.

Good Loan and Bad Loan are project-defined categories based on loan status. They should be explained exactly as defined rather than presented as a universal banking classification.

The dashboard shows associations across dimensions such as purpose, state, term and employment length. These relationships should not be described as causal without additional modelling or controlled analysis.

Average interest rate and average DTI are portfolio-level descriptive measures. They do not replace borrower-level underwriting or risk scoring.

The dataset represents a particular historical period, so conclusions should be validated against newer data before operational decisions are made.

A production environment would also require documented refresh schedules, access controls, data-quality monitoring, lineage, and governance.

22. Interview Explanation — STAR Method

S — Situation

I worked on a finance-domain Bank Loan Analysis project where the requirement was to create a comprehensive report for monitoring lending activity, portfolio health and changes over time. The business needed visibility into applications, funding, collections, interest rate, DTI and Good versus Bad Loan performance.

T — Task

My responsibility was to analyze the loan dataset, translate the business requirements into measurable KPIs, perform SQL analysis, build Power BI dashboards and validate the reported results.

A — Action

I started by profiling and analyzing the data in Python and checking the structure, data types and descriptive statistics. I then used SQL Server for business aggregations and segmentation. In Power BI, I created KPI measures, time-based analysis, Good/Bad Loan analysis and three report pages: Summary, Overview and Details. I also compared SQL results with Power BI outputs to validate the dashboard.

R — Result

The final solution provided an executive Summary view, an analytical Overview and a detailed record-level page. The documented analysis identified 33,243 Good Loan applications at 86.18% and 5,333 Bad Loan applications at 13.82%, giving stakeholders a clear portfolio-quality view.

23. Interview Questions and Strong Answers

Q1. Why did you use Python, SQL and Power BI together?

Answer: Python was useful for initial data exploration and preparation, SQL Server for structured business analysis and validation, and Power BI for interactive reporting and stakeholder consumption. Each tool served a different layer of the workflow.

Q2. How did you define Good and Bad Loans?

Answer: In this project, Good Loans are Fully Paid and Current, while Bad Loans are Charged Off. I used these exact business definitions consistently across Python, SQL and Power BI.

Q3. How did you validate your dashboard?

Answer: I reproduced key dashboard metrics in SQL and compared them with Power BI. I also tested filter contexts, such as Grade A, by applying equivalent SQL filters.

Q4. What is DTI?

Answer: DTI is debt-to-income ratio. It provides a view of debt burden relative to income; the domain documentation notes that lower DTI generally indicates better repayment capacity.

Q5. What is the difference between MTD and MoM?

Answer: MTD measures activity from the beginning of the current month through the selected date. MoM compares one month's result with the previous month and is usually expressed as a percentage change.

Q6. What would you improve next?

Answer: I would add automated data-quality checks, a refresh process, deeper cohort analysis and, if business data allowed, predictive risk modelling with proper validation.

24. Final Project Walkthrough and 60-Second Pitch

60-SECOND INTERVIEW PITCH

“I developed an end-to-end Bank Loan Analysis project in the finance domain using Python, MS SQL Server and Power BI. I started by profiling and analyzing 38,576 loan records with 24 columns. I translated the business requirements into KPIs covering loan applications, funded amount, amount received, average interest rate, DTI, MTD and MoM performance. I then used SQL to perform business analysis and independently validate the results. In Power BI, I built three dashboards: Summary for executive KPIs and Good versus Bad Loans, Overview for monthly, geographic and borrower segmentation, and Details for record-level investigation. The analysis identified 33,243 Good Loan applications, or 86.18%, and 5,333 Bad Loan applications, or 13.82%. The key value was turning raw loan data into a validated, interactive reporting solution that supports portfolio monitoring and data-driven decision-making.”

Final checklist before an interview

• Know the business problem before explaining the tools.

• Explain every KPI in plain business language.

• Know the Good/Bad Loan definition exactly.

• Know the major project numbers and how they were calculated.

• Be able to explain one SQL query end-to-end.

• Be able to explain one Power BI measure and one dashboard visual.

• Explain how you validated Power BI against SQL.

• Do not claim predictive modelling if you only performed descriptive analysis.

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End of Report